

AMD (64-bit) architecture. Do not forget to crosscheck the SHA256 checksum to verify if it's the correct download. Create a CD from the downloaded ISO, select the required hardware and proceed to install.

Recommended hardware

Selection of hardware depends on required throughput and the various features to be used such as VPN, captive portal, additional packages, etc.

For pfSense installation, the computer system must have at least two Ethernet cards — one per WAN and LAN interface. For extra segments such as DMZ/secondary WAN, additional Ethernet card(s) are required.

For writing this series of articles, I have used a system with three Ethernet cards.

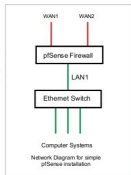


Figure 1: Simple network diagram

Installation

Make sure that data from the hard disk being used for pfSense installation is backed up before proceeding. The installation will delete everything from the disk and create new suitable partitions.

WAN

General Configuration

Enable

- Enable interface

Description

WAN

Enter a description (name) for the interface here.

IPv4 Configuration Type

PPPoE

IPv6 Configuration Type

None

MAC controls

xxxxxxxxxxxx

This field can be used to modify ("spoof") the MAC address of this interface. Enter a MAC address in the following format: xxxxxxxxxx or leave blank

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 (TCP/IP header size) will be in effect.

Figure 2: General configuration

PPPoE Configuration

Username

Password

Confirm

Service name

This field can usually be left empty

Dial on demand

- Enable Dial-On-Demand mode

Idle timeout

If no qualifying outgoing packets are transmitted for the specified number of seconds, the connection is brought down. An idle timeout of zero disables this feature.

Periodic reset

Disabled

Select a reset timing type

Advanced and MLPPP

Advanced and MLPPP

Click for additional PPPoE configuration options. Save first if changes have been made.

Figure 3: PPPoE settings

192.168.33.138 / status_logs.php

May 3 01:23:56	ppp	[wan] IPCP: rec'd Configure Nak #2 (Ack-Sent)
May 3 04:23:56	ppp	[wan] IPADDR 61.2.50.180
May 3 04:23:56	ppp	[wan] 61.2.50.180 is OK
May 3 01:23:56	ppp	[wan] PRIDNS 218.248.255.209
May 3 04:23:56	ppp	[wan] SECDNS 218.248.255.213
May 3 04:23:56	ppp	[wan] IPCP: SendConfigReq #3
May 3 04:23:56	ppp	[wan] IPADDR 61.2.50.180
May 3 04:23:56	ppp	[wan] PRIDNS 218.248.255.209
May 3 04:23:56	ppp	[wan] SECDNS 218.248.255.213
May 3 04:23:56	ppp	[wan] IPCP: rec'd Configure Ack #3 (Ack-Sent)
May 3 04:23:56	ppp	[wan] IPADDR 61.2.50.180
May 3 04:23:56	ppp	[wan] PRIDNS 218.248.255.209
May 3 04:23:56	ppp	[wan] SECDNS 218.248.255.213
May 3 04:23:56	ppp	[wan] IPCP: state change Ack-Sent -> Opened
May 3 04:23:56	ppp	[wan] LayerUp
May 3 04:23:56	ppp	[wan] 61.2.50.180 -> 61.2.48.1
May 3 04:23:56	check_reload_status	Rewriting resolv.conf
May 3 01:23:57	check_reload_status	rc.newwanip starting pppoe0
May 3 04:23:57	ppp	[wan] IFACE: Up event
May 3 04:23:57	ppp	[wan] IFACE: Rename interface ng0 to pppoe0
May 3 04:23:58	ppp-fpm	19369 /rc.newwanip: rc.newwanip: info: starting on pppoe0.

Figure 4: PPPoE logs